

WASP-1b

R-2043

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-1_121212 · created 2026-08-08T23:22:57+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456273.7127 +/- 0.0046 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.037 +/- 0.02
Transit depth (Rp/Rs)^2	0.14 +/- 0.15 [%]
Orbital Inclination (inc)	89.76 +/- 1.42
Airmass coefficient 1 (a1)	1.218 +/- 0.012
Airmass coefficient 2 (a2)	-0.0941 +/- 0.0028
Scatter in the residuals of the lightcurve fit is	0.91 %
Best Comparison Star	None
Optimal Aperture	4.44
Optimal Annulus	11.16
Transit Duration (day)	0.144 +/- 0.0032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.037 ± 0.02 vs 0.1036 (deviation 2.32σ)
Duration fit vs published	0.144 d vs 0.1567 d (deviation 2.75σ)
Mid-time O–C	11.6 min
Depth SNR	1.3
Residual scatter	0.91 %

Light curve

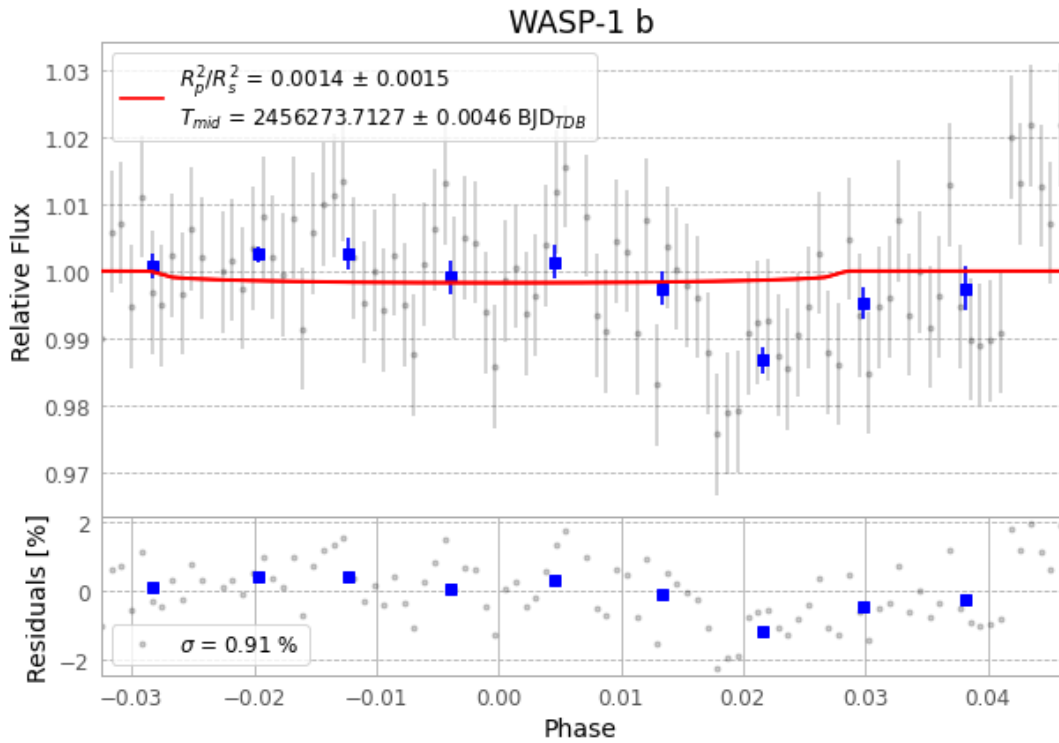


Figure 1 — FinalLightCurve_WASP-1 b_2012-12-11.png (SHA-256 c0a3e2c21aead779...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [349, 286], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d5f94139a147d3c3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

98 FITS frames (64 MB), WASP-1121212025713.FITS → WASP-1121212074812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-121212.log (c222bc79e0cd...), FinalLightCurve_WASP-1 b_2012-12-11.png (c0a3e2c21aea...), FinalLightCurve_WASP-1 b_2012-12-11.csv (254285f7ffc0...)

Compute resources

Wall time 252.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 23:23Z.